Electric Vehicle Market Performance & Insights

Project Proposal

Repository: https://github.com/manasgite54-cmd/EV-project

Purpose and Context

Electric Vehicle Market Performance & Insights combines data engineering and descriptive analytics. Python and Pandas prepare the records, SQLite/SQL support structured storage and queries, Tableau Public provides interactive visual analysis, and Flask with HTML/CSS provides a simple multi-page user interface.

Detailed Project-Specific Content

- Problem statement: How can EV records be cleaned, queried and visualized so users can compare available vehicle attributes while avoiding unsupported conclusions about sales or market share?
- Functional scope: read and validate source data; apply documented cleaning rules; export the processed CSV and JSON report; run descriptive comparisons; store/query records using SQLite/SQL; provide Tableau dashboard and story; and expose Overview, Dashboard, Story and Methodology pages.
- Non-functional needs: reproducibility, readable charts, clear units, reliable file paths, maintainable code, responsive layout, graceful handling of unavailable embeds and no committed credentials.
- Technology rationale: Pandas is suitable for tabular preparation; SQLite is lightweight for a local relational database; Tableau supports interactive exploration; Flask connects pages and embedded visualizations without requiring a complex frontend framework.
- Deliverables: repository, processed dataset, cleaning report, database/SQL assets, visualization links, web interface, methodology, testing evidence and presentation script.
- Out of scope unless separately implemented: live data feeds, user accounts, online transactions, predictive modeling, causal inference, verified sales-volume metrics and production deployment.
- Acceptance: the CSV/report are readable and consistent; SQL queries work; dashboard/story links open; Flask routes render; and all conclusions are traceable to fields and calculations.

Project References and Artifact Locations

Project: Electric Vehicle Market Performance & Insights Repository: https://github.com/manasgite54-cmd/EV-project Processed dataset: data/processed/ev_market_clean.csv Reported processed records: 299,999 Cleaning report: data/processed/cleaning_report.json Tableau Dashboard: https://public.tableau.com/views/Book1_17931030431560/EVMarketOverview?:showVizHome=no&:embed=yes Tableau Story: https://public.tableau.com/views/Book1_17931030431560/Story1?:showVizHome=no&:embed=yes Accuracy note: the record count and artifact paths above are reported by the project owner. Exact schema, source provenance, units, calculated statistics, test results, and deployment status must be checked against the repository and actual artifacts. No unsupported results, approvals, or signatures are claimed.

Status: Detailed project-specific document. Any [TO BE PROVIDED], [TO BE VERIFIED] or [TO BE CALCULATED] item requires confirmation. No supervisor approval, signature, unverified test pass or successful deployment is implied.